Name 1 2 3 4 5

A1

Claim: Let V be a vector space with basis $\{e_i\}$. Then $V^* = \{\ell \mid \ell \text{ is a linear map } V \to \mathbb{R}\}$ is also a vector space with basis $\{\vartheta_j\}$, where the ϑ_j are defined by $\vartheta_j(e_i) = \delta_{ij}$.

proof: Let $\ell \in V^*$. By definition, there are $\alpha_i \in \mathbb{R}$, such that $\ell(e_i) = \alpha_i$. It follows that $\ell = \sum_j \alpha_j \vartheta_j$. Indeed:

$$\ell(v) = \ell\left(\sum_i v_i e_i\right) = \sum_i v_i \ell(e_i) = \sum_i v_i \alpha_i$$

$$= \sum_{ij} v_i \alpha_j \delta_{ij} = \sum_{ij} v_i \alpha_j \vartheta_j(e_i)$$

$$= \sum_j \alpha_j \vartheta_j\left(\sum_i v_i e_i\right)$$

$$= \left(\sum_j \alpha_j \vartheta_j\right)(v) \qquad \forall v \in V$$

$$\Rightarrow \ell = \sum_j \alpha_j \vartheta_j. \qquad \blacksquare$$